

SIMATS ENGINEERING
CO3- ASSESSMENT TOOL 2
SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING

COURSE CODE: ECA09

Code of Conduct:

I L. Vivekananda Reddy (192412380) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Q11. Periodogram-Based Spectral Estimation

Simulate a noisy sinusoidal signal and estimate its power spectral density using the periodogram method. (25) (BL3) (WK4) Given Data:

- Signal Frequencies = 200 Hz and 500 Hz
- Sampling Frequency = 2 kHz
- Number of Samples = 2048
- SNR = 15 dB Tasks:

Aim

To estimate the Power Spectral Density (PSD) of a noisy sinusoidal signal using the Periodogram method in MATLAB and identify the dominant frequency components.

Apparatus / Software Required

- MATLAB
- Computer/Laptop
- MATLAB Signal Processing Toolbox

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Parameter	Value
Signal frequencies	200 Hz, 500 Hz
Sampling frequency	2 kHz
Number of samples	2048
SNR	15 dB
Method	Periodogram

Procedure

1. Open MATLAB and create a new script.
2. Set the sampling frequency to **2 kHz** and number of samples to **2048**.
3. Generate two sinusoidal signals at **200 Hz and 500 Hz**.
4. Add Gaussian noise to obtain an **SNR of 15 dB**.
5. Apply the MATLAB periodogram() function to the noisy signal.
6. Plot the noisy time-domain signal.
7. Plot the estimated **Power Spectral Density (PSD)**.
8. Use findpeaks() to identify the dominant frequency components.
9. Observe the peaks around **200 Hz and 500 Hz**

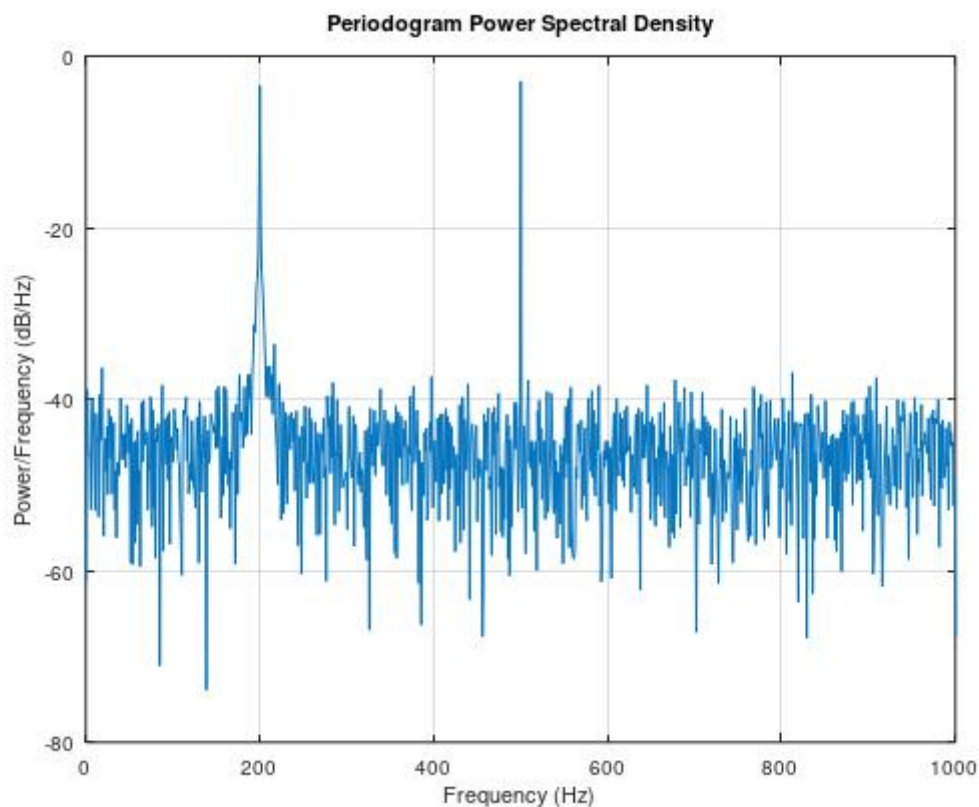
Program

```

1  clc;
2  clear;
3  close all;
4
5  % Parameters
6  Fs = 2000;          % Sampling frequency
7  N = 2048;          % Number of samples
8  f1 = 200;           % First frequency
9  f2 = 500;           % Second frequency
10 SNR = 15;           % SNR in dB
11
12 % Time vector
13 t = (0:N-1)/Fs;
14
15 % Generate clean signal
16 x = sin(2*pi*f1*t) + sin(2*pi*f2*t);
17
18 % Calculate signal power
19 signal_power = mean(x.^2);
20
21 % Calculate noise power for 15 dB SNR
22 noise_power = signal_power / (10^(SNR/10));
23
24 % Generate Gaussian noise
25 randn("seed",1);
26 noise = sqrt(noise_power) * randn(size(x));
27
28 % Noisy signal
29 x_noisy = x + noise;
30
31 % Periodogram
32 [Pxx, f] = periodogram(x_noisy, [], N, Fs);
33
34 % Plot noisy signal

```

Output



Result

Thus, the PSD of the noisy sinusoidal signal was successfully estimated using the Periodogram method. The dominant frequency components were identified at approximately 200 Hz and 500 Hz, demonstrating the ability of the periodogram to estimate the frequency content of a noisy signal.